

Gavin Bowden

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EDUCATION

Purdue University, West Lafayette Indiana

Aug. 2023 – Aug. 2026

Bachelor of Science in Computer Science & Artificial Intelligence, English Minor

Course Highlights: Natural Language Processing; Machine Learning; Data Mining; Analysis of Algorithms; Human–Computer Interaction

EXPERIENCE

Langley Student Volunteer Program – **NASA Langley Research Center**, Hampton, VA

Jun. 2025 – Aug. 2025

- Migrated NASA Langley's mail routing system from legacy Oracle APEX to a Microsoft Power App on a Microsoft Lists backend, serving 3,400+ staff across 190 buildings
- Redesigned the mailstop data model and search interface so staff could look up and update assignments directly, without going through a system administrator

ML Based Cooling Tower Predictive Maintenance

- Merged a Langley AFB local-weather dataset with cooling-tower maintenance logs to engineer stronger time-series features
- Built and evaluated a boosted decision tree model to predict cooling tower maintenance needs, tuned with cross-validation and Bayesian hyperparameter search
- Presented findings to 25+ NASA engineers and staff, initiating collaboration to further refine the model

Student Researcher – **Howmet Aerospace** / Purdue Data Mine

Aug. 2023 – May 2024

- Developed CNN-based computer vision models to detect surface defects in aircraft jet turbine blades, and presented results at the Data Mine Symposium

PROJECTS

Inkblot – Local-First AI Writing Notebook (Rust / Tauri, React, ONNX Runtime)

Feb. 2026 – Present

- A desktop writing app for creatives with a spatial canvas where every AI feature runs on-device, so user text never leaves the machine
- Wrote [inkblot-dictation](#), an open-source Rust crate for offline speech-to-text: live mic capture, voice-activity chunking, and streaming transcripts from a quantized Whisper model on Metal GPU
- Added local read-aloud (Kokoro TTS) and in-editor synonym suggestions, both served through ONNX Runtime in Rust

Context-Aware Synonym Generation Pipeline (Python, PyTorch, spaCy) [g3vin/lexical-substitution](#)

Apr. – May 2026

- Designed a multi-stage lexical substitution pipeline – FLAN-T5/WordNet candidate generation, spaCy filtering, Sentence-BERT scoring, cross-encoder reranking
- Fine-tuned a RoBERTa substitute-quality classifier and trained a learned ranker, reaching 0.53 NDCG@10 and 0.58 precision@1 on the SWORDS benchmark

Event Pass – Live Attendance & Raffle System (React, Firebase) [BoilerBookClub/number-caller](#)

Apr. 2026

- Built the real-time check-in and number-calling platform Boiler Book Club runs its giveaways on, syncing a projected room display, a staff control panel, and 100+ concurrent attendee phones over Firestore listeners
- Implemented Discord OAuth against Firebase Auth custom tokens minted in Cloud Functions, with rotating 60-second check-in QR codes and per-attendee claim codes redeemed server-side to block duplicate claims
- Modeled multi-round events with auto group advancement and backlog limits, enforced by Firestore rules and deployed via GitHub Actions

LEADERSHIP AND INVOLVEMENT

President – **Boiler Book Club**

May 2023 – April 2026

- Led a community of 1,000+ readers and organized in-person events with 300+ attendees
- Proposed, secured, and managed multi-thousand-dollar budgets for large-scale events

Treasurer – **Purdue Archery Club**

May 2023 – April 2026

- Managed club finances and tournament funding
- Built [purduearchery.club](#) on Firebase, using Discord OAuth as the club's identity source of truth for QR practice check-in, pass sales tracking, and an officer CMS for the public site

SKILLS

Languages: Python, Java, C, C++, Rust, JavaScript, SQL, x86 ASM

ML / Data: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, ONNX Runtime, spaCy, NumPy, Pandas, OpenCV

Web / App: React, Node.js, Tauri, Vite, REST APIs, HTML/CSS, Firebase (Firestore, Auth, Cloud Functions), SQLite, Oracle APEX

Tools: Git/GitHub, GitHub Actions CI/CD, Linux/Bash, Microsoft Power Platform, Agile/Scrum